# FBISE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

# MATHEMATICS

Past Paper 2022 • Paper Code: 2HA-I 2211 — D(8111)

Retyped faithfully from the original board paper — same questions, no changes.

Note: MATHEMATICS HSSC–II. SECTION – A (Marks 20). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 8111. Fill the relevant bubble against each question:

Q1. $x = a\cos\theta$, $y = b\sin\theta$ are parametric equations of:

(A) Circle
(B) Parabola
(C) Ellipse
(D) Hyperbola

Q2. Which of the following represents $f^{-1}(5)$ if $f(x) = x^{(1/3)} + 2$

(A) 1
(B) 3
(C) 9
(D) 27

Q3. In which of the following intervals, $f(x) = 4x - 2x^2$ is increasing?

(A) $0 \le x \le 1$
(B) $0 < x < 1$
(C) $1 \le x \le 2$
(D) $0 < x < 2$

Q4. What result will occur, in evaluating $\lim_{n\to\infty}(1 + 2/n)^{3n}$

(A) $e$
(B) $e^2$
(C) $e^3$
(D) $e^6$

Q5. For a function $f(x) = a\sin 3x$ and $f''(\pi/3) = 6$ then what is the value of a?

(A) $-2$
(B) 2
(C) 3
(D) 6

Q6. $d/dx(\sec^{-1}x + \operatorname{cosec}^{-1}x) =$

(A) $-1$
(B) 0
(C) 1
(D) 2

Q7. $\int(1/\sqrt{x})(dx/(\sqrt{x}+1)) =$

(A) $-\ln(\sqrt{x}+1)+c$ (B) $\ln(\sqrt{x}+1)+c$

(C) $2\ln(\sqrt{x}+1)+c$ (D) $-2\ln(\sqrt{x}+1)+c$

Q8. Which one of the following results occurs of the integral $\int_0^2 dx/(x^2+4)$

(A) $\pi/6$ (B) $\pi/8$

(C) $\pi/4$ (D) $\pi/2$

Q9. If $\int_0^2 f(x)dx = 3$ then what is the value of k if $\int_0^2(3f(x)+4)dx = k$

(A) 11 (B) 17

(C) 20 (D) 23

Q10. The points A(2,5) and B(3,−2) are the ends of a diameter of a circle, what is the radius of a circle?

(A) $2\sqrt{5}$ (B) $5\sqrt{2}$

(C) $5/\sqrt{2}$ (D) $2/\sqrt{5}$

Q11. A line cuts the x-axis at (2,0) and y-axis at (0,−4), then equation of a line is:

(A) $2x-y-4 = 0$ (B) $2x-y+4 = 0$

(C) $x+2y-4 = 0$ (D) $x-2y+4 = 0$

Q12. Pair of lines represented by Homogeneous equation $ax^2 + 2hxy + by^2 = 0$ through origin will be real and coincident if:

(A) $h^2 > ab$ (B) $h^2 < ab$

(C) $h^2 = ab$ (D) $a+b = 0$

Q13. The solution set of $2y+5 > 4y-3$

(A) $y > -4$ (B) $y > 8$

(C) $y < -4$ (D) $y < 4$

Q14. The line $y = mx + c$ will be tangent to a circle $x^2 + y^2 = a^2$ if:

(A) $c = \pm m\sqrt{(1+a^2)}$ (B) $c = \pm a\sqrt{(1+m^2)}$

(C) $c = \pm m\sqrt{(1-a^2)}$ (D) $c = \pm a\sqrt{(1-m^2)}$

Q15. What is the Length of Latus Rectum of Parabola $x^2 = 5y$

(A) 5 (B) 20

(C) 5/4 (D) 10

Q16. Which one of the following represents the graph of $9x^2-18x+4y^2+8y-23 = 0$?

(A) Circle (B) Parabola

(C) Ellipse (D) Hyperbola

Q17. The co-vertices of hyperbola $x^2/16 - y^2/4 = 1$ are:

(A) $(0,\pm4)$ (B) $(\pm2,0)$

(C) $(\pm4,0)$ (D) $(0,\pm2)$

Q18. The area of the triangle whose adjacent sides are $3\bar{i}+4\bar{j}$ and $12\bar{i}+9\bar{j}$ is:

(A) 45/2 (B) 21/2

(C) 55/2 (D) 25/2

Q19. If vectors $v = \bar{i}-3\bar{j}+4\bar{k}$ and $w = \lambda\bar{i}+9\bar{j}-12\bar{k}$ are parallel then what is the value of $\lambda$?

(A) $-3$ (B) 3

(C) $-9$ (D) 9

Q20. What is the volume of a parallelepiped if its edges are $2\bar{i}-4\bar{j}+5\bar{k}$, $2\bar{i}-3\bar{j}+6\bar{k}$, $-\bar{j}-\bar{k}$?

(A) 0 (B) 3

(C) 15 (D) 24

> Note: —2HA-I 2211-8111 (HA)—. Page 1 of 2. Page 2 of 2.

> Note: MATHEMATICS HSSC–II. Time allowed: 2:35 Hours. Total Marks Sections B and C: 80. NOTE: Attempt any twelve parts from Section 'B' and any four questions from Section 'C' on the separately provided answer book. Use supplementary answer sheet i.e. Sheet-B if required. Write your answers neatly and legibly. Graph paper will be provided on Demand.

## Q. SECTION – B (Marks 48)

Q. 2 Attempt any TWELVE parts. All parts carry equal marks. (12 x 4 = 48)

(i) Let the real valued function, f and g defined by $f(x) = 4x+1$ and $g(x) = 2x^2+5x$ obtain the expression for: a. f((g)) b. g(f(x)) c. f(f(x)) d. g(f(x))

(ii) Evaluate $\lim_{x\to0} (\sqrt{(x+5)} - \sqrt{5})/x$

(iii) Find dy/dx if $x = 3at/(1+t^3)$, $y = 3at^2/(1+t^3)$

(iv) If $y = \tan(4\tan^{-1}(x/4))$ show that $dy/dx = 16(1+y^2)/(16+x^2)$

(v) Use implicit rule to find the second derivative of the function $y = x + \tan^{-1}y$

(vi) If $x = \cos\theta$; $y = \cos n\theta$ show that $(1-x^2)y_2 - xy_1 + n^2y = 0$

(vii) Find the area between the x-axis and the curve $f(x) = x^3 - 2x$ from $x = 0$ to $x = 3$

(viii) Evaluate $\int x^3\sqrt{(1+x^2)}\,dx$

(ix) Find the point two-fifth of the way along the line segment A(−3,5) to B(5,3).

(x) Find the angle $\theta$ form the lines $L_1$ and $L_2$: $L_1$: $7x+3y-9 = 0$; $L_2$: $5x-2y+2 = 0$

(xi) Graph the feasible solution region of the system of linear inequalities by shading, also find the corner points. $3x+7y \leq 21$, $x-y \leq 3$, $x \geq 0$, $y \geq 0$

(xii) Find the equation of parabola with focus (1,3) and vertex (4,3).

(xiii) Find the equation of parabola$_1$, with Directrix, $y = 3$ and vertex (2,2).

(xiv) Write the equation of ellipse with vertices at $(-1,2)$ and $(7,2)$ and 2 is the length of semi minor axis whereas major axis is horizontal.

(xv) Prove that $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

(xvi) Find constant $\alpha$ so that vectors are coplaner: $\bar{i}-\alpha\bar{j}-\bar{k}$, $\bar{i}+\bar{j}+2\bar{k}$ and $\alpha\bar{i}-\bar{j}+\bar{k}$

## Q. SECTION – C (Marks 32)

Note: Attempt any FOUR questions. All questions carry equal marks. (4 x 8 = 32)

Q. 3 If $\theta$ is measured in radian then prove that $\lim_{\theta\to 0} \sin\theta/\theta = 1$. a. Draw the figure and give explanation. b. Find area of triangles in figure. c. From figure, see the inequalities of area and prove the theorem.

Q. 4 Consider the function $f(x) = \sin x + (1/\sqrt{2})\cos 2x$ where $x \in (0,2\pi)$. find the extreme values of the functions in the interval $x \in (0,2\pi)$. a. Find function $f'(x)$ b. Find $f''(x)$ c. Find the values of $x \in (0,2\pi)$ for which $f(x)$ has maximum or minimum values d. Find possible extreme values of $f(x)$

Q. 5 Integrate $\int(2x+5)/((x-3)^2(x^2-x+5))dx$. a. Resolve $(2x+5)/((x-3)^2(x^2-x+5))$ into Partial fraction b. After Partial Fraction Integrate the result $\int(2x+5)/((x-3)^2(x^2-x+5))dx$

Q. 6 The diagram shows two Lines $L_1$ and $L_2$ passing through points: $L_1$: joins A(2,7) and B(7,10); $L_2$: joins C(1,1) and D$(-5,3)$. a. Find the slope of lines $L_1$ and $L_2$ b. Find the angle between the lines $L_1$ and $L_2$ c. Find the equations of line $L_1$ and $L_2$ d. Find the point of contact where line $L_1$ and $L_2$ intersect. (Note: a diagram of the lines is given.)

Q. 7 Find the maximum and minimum values of f and g defined as $f(x) = 3x+5y$ and $g(x) = 6x+8y$ under the constraints. $2x-3y \leq 6$, $2x+y \geq 2$, $2x+3y \leq 12$, $x \geq 0$, $y \geq 0$

Q. 8 Find the equations of tangent and normal lines at a point (3,12/5) to ellipse $x^2/25 + y^2/9 = 1$. For what value of C the line $x+y+c = 0$ will touch the ellipse $x^2/25 + y^2/9 = 1$

> Note: —2HA 2211 (HA)—. Page 1 of 2 (Mathematics). Page 2 of 2 (Mathematics).